INTERACTION DESIGN SPECIFICATION

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1. Purpose & Scope

This Interaction Design Specification (IDS) defines the complete interactive behaviour of two feature areas within the portfolio modernization project: the Project Grid and the Image Lightbox. It covers component states, transitions, animation parameters, keyboard and touch interactions, ARIA semantics, and error/edge case handling.

This document is derived from PRD_Project_Gallery_and_Lightbox.docx and extends it with interaction-layer detail not captured in requirements format. It is the authoritative reference for implementation and QA validation of all interactive behaviour.

1.1 Components Covered

- ProjectGrid — responsive project listing container
- ProjectCard — individual project tile with thumbnail, title, tags, and hover behaviour
- ProjectDetail — inline project display with 5 layout variants
- ProjectGallery — thumbnail grid linking to lightbox
- Lightbox — full-screen modal image viewer
- LoadingSkeleton — shimmer placeholder during data fetch

1.2 Out of Scope

- Tag filtering and keyword search (future Enhancement 1)
- Individual project detail routes /projects/[id] (future Enhancement 5)
- Pinch-to-zoom in lightbox (optional future feature)

2. Design Tokens & Motion Principles

2.1 Animation Timing

All durations and easing values are defined as design tokens and applied consistently across components.

TOKEN	VALUE	EASING	USAGE
`duration.instant`	`0ms`	—	Reduced-motion fallback
`duration.fast`	`150ms`	ease-in-out	Button press, active states
`duration.standard`	`200ms`	ease-out	Hover transitions, focus rings
`duration.enter`	`300ms`	ease-out	Lightbox open, fade-in entrance
`duration.exit`	`200ms`	ease-in	Lightbox close
`duration.slide`	`250ms`	cubic-bezier(0.4, 0, 0.2, 1)	Lightbox image navigation
`duration.scroll`	`400ms`	ease-out	Scroll-triggered fade-in
`duration.shimmer`	`1500ms`	linear, infinite	Skeleton loading shimmer

2.2 Reduced Motion Principle

All animations and transitions are disabled when the user has set prefers-reduced-motion: reduce at the OS level. This applies globally via a CSS media query override and is additionally enforced in JavaScript via the useReducedMotion() hook.

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    transition-duration: 0.01ms !important;
  }
}
```

> ℹ️ *Reduced motion must not remove functionality — all interactions remain available, only visual motion is suppressed. State changes (hover styles, focus rings, open/close) still apply instantly.*

2.3 Keyframe Definitions

KEYFRAME	TRANSFORM	USAGE
`fadeIn`	opacity: 0 → 1, translateY: 20px → 0	Lightbox open, scroll entrance
`scaleIn`	opacity: 0 → 1, scale: 0.95 → 1	Modal entrance
`slideLeft`	translateX: 0 → -100%	Lightbox: advance to next image
`slideRight`	translateX: 0 → 100%	Lightbox: back to previous image
`slideInLeft`	translateX: 100% → 0	Lightbox: new image enters from right
`slideInRight`	translateX: -100% → 0	Lightbox: new image enters from left
`shimmer`	background-position: -1000px → 1000px	Skeleton loading
`pulse`	opacity: 1 → 0.5 → 1 (infinite)	Loading spinner fallback

3. ProjectCard Component

3.1 Component States

ProjectCard has four interactive states. State transitions use duration.standard (200ms ease-out) unless noted.

STATE	TRIGGER	VISUAL CHANGE	REDUCED MOTION
Rest	Default	elevation: 1, border-radius: 8px, scale: 1.0	Same — no change
Hover	Mouse enter	transform: scale(1.02), elevation shadow: 4, 200ms ease-out	Instant: elevation 4 only, no scale
Focus	Keyboard Tab	2px solid outline, outlineOffset: 2px, 3:1 contrast min	Same — focus ring always visible
Active	Mouse/key press	transform: scale(0.99), 150ms ease-in-out	Instant: no scale

3.2 Responsive Dimensions

BREAKPOINT	VIEWPORT	IMAGE HEIGHT	TITLE SIZE	TAGS SHOWN
xs	< 600px	200px	1.1rem	2 + overflow
sm	600–899px	180px	1.15rem	3 + overflow
md	900–1199px	200px	1.2rem	3 + overflow
lg	≥ 1200px	220px	1.25rem	4 + overflow

3.3 Keyboard Interaction

KEY	ACTION	NOTES
`Tab`	Focus next card	Focus ring must be visible at all times
`Shift+Tab`	Focus previous card	
`Enter`	Activate card	Same as click
`Space`	Activate card	Same as Enter

3.4 Error & Edge Cases

- Missing thumbnail: render a placeholder with project initials or a generic icon at the same aspect ratio
- No tags: tag row is omitted; card height adjusts gracefully without empty space
- Long title: truncate at 2 lines with ellipsis; full title available via title attribute for screen readers
- Empty projects array: ProjectGrid renders an accessible empty state message, not a blank page

4. ProjectGrid Component

4.1 Layout Behaviour

BREAKPOINT	COLUMNS	GAP	NOTES
xs / sm (< 900px)	1	24px	Full-width cards
md (900–1199px)	2	24px	
lg (≥ 1200px)	3	24px	Maximum density

4.2 Loading State

While project data is fetching, ProjectGrid renders skeleton placeholders that match the exact dimensions of real cards at each breakpoint. This prevents layout shift when content loads.

- Skeleton count matches the expected number of visible cards (6 on desktop, 4 on tablet, 2 on mobile)
- Shimmer animation runs at 1500ms linear infinite
- Shimmer disabled under prefers-reduced-motion; static gray fill (#E0E0E0) used instead
- aria-busy="true" set on the grid container while loading
- aria-label="Loading projects" on skeleton container

4.3 Scroll-Triggered Entrance

Project cards fade in as they enter the viewport using the useScrollAnimation hook with IntersectionObserver.

PARAMETER	VALUE
Intersection threshold	0.1 (10% of card visible triggers animation)
Animation	fadeIn: opacity 0→1, translateY 20px→0, 400ms ease-out
Stagger delay	None — cards animate independently on viewport entry
Reduced motion	Cards are immediately visible; no animation plays
Once only	Animation fires once per card; does not repeat on scroll-out

5. Project Detail — Layout Variants

ProjectDetail renders one of five layout variants determined at runtime by viewport width, video presence, and the project's altGrid flag.

VARIANT	CONDITION	VIEWPORT	STRUCTURE
`wide-video`	video present	≥ 900px	Full-width title / Left 1/3: tags + description / Right 2/3: video + 4-col thumbnail grid
`wide-regular`	no video, altGrid: false	≥ 900px	Full-width title / Left 2/3: description (tags float right) / Right 1/3: 2-col thumbnail grid
`wide-alternate`	no video, altGrid: true	≥ 900px	Full-width title / Full-width description / 4-col thumbnail grid below
`narrow-video`	video present	< 900px	Stacked: title / video (16:9) / tags / description / 4-col thumbnail grid
`narrow`	no video	< 900px	Stacked: title / tags / description / 4-col thumbnail grid

5.1 Thumbnail Grid Specifications

LAYOUT VARIANT	COLUMNS	GAP	THUMBNAIL SIZE	NOTES
`wide-video`	4	12px	~150px	
`wide-regular`	2	16px	~180px	
`wide-alternate`	4	12px	~150px	
`narrow (all)`	4	8px	~80px	Mobile-optimised density

6. Lightbox Component

The Lightbox is a full-screen modal image viewer. It is the most interaction-complex component in the system and has strict accessibility requirements as a WAI-ARIA dialog pattern.

6.1 Component States

STATE	TRIGGER	BEHAVIOUR	REDUCED MOTION
Closed	Default / after close	Not mounted; no DOM presence	Same
Opening	Thumbnail click or Enter/Space	Overlay fades in (300ms ease-out), image scales in from 0.95 → 1	Instant: no transition
Open / Idle	After open transition	Image visible, nav buttons visible, focus trapped	Same
Navigating	Arrow key / button / swipe	Current image slides out, next image slides in (250ms)	Instant image swap
Closing	Escape / close button	Overlay fades out (200ms ease-in)	Instant: no transition

6.2 Open / Close Transitions

Opening sequence

1. Overlay renders at opacity: 0
2. Body scroll locked: overflow: hidden, position: fixed
3. Focus moved to close button (first focusable element)
4. Overlay fades to opacity: 1 over 300ms ease-out
5. Image scales from 0.95 → 1.0 simultaneously

Closing sequence

6. Overlay fades to opacity: 0 over 200ms ease-in
7. On transition end: component unmounts
8. Body scroll lock released
9. Focus returns to the thumbnail that triggered open

> ℹ️ *Focus return is critical for keyboard and screen reader users. The triggering element must be stored in a ref before opening and restored on close.*

6.3 Image Navigation

Navigation is wrapping: advancing past the last image returns to the first; going back from the first image wraps to the last.

DIRECTION	DURATION	ANIMATION
Next (→)	`250ms`	Current image: translateX 0 → -100%. New image: translateX 100% → 0. Both simultaneous.
Previous (←)	`250ms`	Current image: translateX 0 → 100%. New image: translateX -100% → 0. Both simultaneous.
Reduced motion	`0ms`	Instant swap — no slide animation

6.4 Keyboard Interactions

KEY	ACTION	NOTES
`← Arrow Left`	Previous image	Wraps from first to last
`→ Arrow Right`	Next image	Wraps from last to first
`Escape`	Close lightbox	Fires onClose; focus returns to trigger
`Tab`	Next focusable (trapped)	Cycles: Close → Prev → Next → Close
`Shift+Tab`	Previous focusable (trapped)	Reverse cycle within lightbox only
`Enter / Space`	Activate focused button	On nav buttons or close button

6.5 Touch / Swipe Interactions

Swipe is implemented via the useSwipe hook tracking touchstart, touchmove, and touchend events. Visual swipe-follow (image moves with finger) is enabled when reduced motion is not preferred.

GESTURE	ACTION	THRESHOLD	NOTES
Swipe left	Next image	> 50px horizontal, < 30px vertical	
Swipe right	Previous image	> 50px horizontal, < 30px vertical	
Swipe down	Close lightbox	> 100px vertical	
Snap back	No action	Below threshold	Image returns to position, 150ms ease-out

touch-action: manipulation must be set on the lightbox container to prevent double-tap zoom on iOS Safari.

6.6 Mobile Layout

ELEMENT	SPECIFICATION
Navigation arrows	Bottom centre, 48×48px touch targets, semi-transparent background (rgba 0,0,0,0.4)
Close button	Top-right, 48×48px, always visible
Image counter	Top-left, non-interactive, white text on semi-transparent background
Caption	Bottom, above navigation arrows, max 2 lines with ellipsis overflow
Safe area	env(safe-area-inset-*) applied to padding on notched devices
Body scroll	overflow: hidden + position: fixed on body while open (iOS Safari fix)

6.7 ARIA Semantics

ATTRIBUTE	VALUE & PURPOSE
`role`	"dialog" — identifies the lightbox as a modal dialog
`aria-modal`	"true" — informs screen readers content behind is inert
`aria-label`	"Image gallery lightbox" — names the dialog for screen readers
`aria-describedby`	Points to caption element ID — screen reader reads caption after image alt
`aria-live`	"polite" on counter — announces "3 of 12" on navigation without interrupting
`alt (image)`	ProjectImage caption text — descriptive alt for current image
`aria-label (close)`	"Close lightbox"
`aria-label (prev)`	"Previous image"
`aria-label (next)`	"Next image"

6.8 Focus Management

- On open: focus moves immediately to the close button (first focusable element in DOM order)
- Focus trap: Tab and Shift+Tab cycle only through: Close button → Prev button → Next button → (back to Close)
- On close: focus returns to the exact thumbnail element that triggered the open
- If trigger element is no longer in DOM on close, focus falls back to the main content landmark
- All three interactive elements (close, prev, next) must be reachable and have visible focus rings at 3:1 contrast minimum

6.9 Edge Cases

SCENARIO	EXPECTED BEHAVIOUR
Single image in gallery	Prev/Next buttons hidden or disabled; keyboard arrows do nothing
Image fails to load	Render broken image placeholder with alt text; navigation still functional
Very long caption	Truncate at 2 lines on mobile with ellipsis; full text available on desktop
Rapid keyboard navigation	Queue or debounce: do not start new slide transition until current one completes
Window resize while open	Responsive layout recalculates; image re-centres; no layout break
Browser back button	Lightbox closes; does not navigate away from page
Swipe threshold not met	Image snaps back to origin position over 150ms ease-out

7. LoadingSkeleton Component

Skeleton screens replace real content during data fetch. Their dimensions match the real content layout precisely to prevent cumulative layout shift (CLS).

7.1 Skeleton States

STATE	ANIMATION	REDUCED MOTION
Loading (motion on)	Shimmer: background gradient slides 1500ms linear infinite	—
Loading (motion off)	—	Static gray fill (#E0E0E0); no animation
Transition to content	Skeleton fades out 200ms as real content fades in	Instant swap

7.2 ARIA Requirements

- aria-busy="true" on the grid container while skeletons are visible
- aria-label="Loading projects" on the skeleton wrapper
- Skeleton shapes are aria-hidden="true" — they are decorative
- When content loads: aria-busy removed; screen reader announces actual content

8. WCAG 2.2 Level AA Quick Reference

The following table maps WCAG 2.2 Level AA success criteria to their implementation across the components covered by this specification.

SC	CRITERION	IMPLEMENTATION
1.1.1	Non-text content	All images: alt text from ProjectImage caption; decorative skeletons: aria-hidden
1.4.3	Contrast (text)	4.5:1 minimum; enforced by theme system across light, dark, and high-contrast modes
1.4.11	Non-text contrast	3:1 minimum on focus rings, gallery thumbnail borders, and button icons
2.1.1	Keyboard	All interactions available via keyboard: Tab, Enter, Space, Escape, Arrow keys
2.1.2	No keyboard trap	Lightbox focus trap allows Escape exit at all times
2.3.3	Animation	prefers-reduced-motion disables all animations globally
2.4.3	Focus order	DOM order matches visual order; no tabindex > 0
2.4.7	Focus visible	2px solid outline, outlineOffset: 2px, 3:1 contrast in all themes

2.5.8	Target size	All interactive controls >= 44x44px; lightbox mobile controls >= 48x48px
4.1.2	Name, role, value	ARIA roles, labels, and states on all controls; aria-live on image counter

9. useSwipe Hook Specification

The useSwipe hook encapsulates all touch gesture detection for the Lightbox. It is a standalone, reusable hook that attaches touch event listeners to a provided ref.

```
interface UseSwipeOptions {
  onSwipeLeft?:  () => void;   // Next image
  onSwipeRight?: () => void;   // Previous image
  onSwipeDown?:  () => void;   // Close lightbox
  threshold?:    number;       // px — default 50 horizontal, 100 vertical
  enabled?:      boolean;      // default true
}
```

```
function useSwipe(
  ref: RefObject<HTMLElement>,
  options: UseSwipeOptions
): void
```

9.1 Detection Logic

10. Record touch start coordinates on touchstart
11. Track delta on touchmove
12. On touchend: calculate deltaX and deltaY
13. If |deltaX| > threshold AND |deltaY| < 30px → horizontal swipe
14. If deltaY > 100px AND |deltaX| < 30px → downward swipe
15. Fire corresponding callback; ignore if below threshold

The vertical constraint (< 30px) on horizontal swipes prevents accidental navigation when the user is scrolling the page vertically.

10. Document Change Log

VERSION	DATE	AUTHOR	CHANGES
1.0	March 2026	Sing Chan	Initial document — derived from PRD_Project_Grid_and_Lightbox.docx with interaction-layer extensions